



Booming Boom jets

Boom's futuristic supersonic passenger jet gets 76 orders for its new XB-1 design which yet to be demonstrated. <http://dgit.in/boomBoom>



Autonomous future for Honda

Pushed by supercars and robots, they plan to introduce level 4 autonomous driving capability by 2025, hopefully. <http://dgit.in/FondueH>

which is done via a technology called CRISPR, renders the embryo incapable of growing a certain organ. Then, when the embryo develops, the human stem cells step in to grow that organ. The resulting organism is a mixture of both human and animal. Such a combination of two (or more) organisms is scientifically known as a chimera. The process also allows for other studies such as figuring out growth patterns and disease development to get valuable data that wouldn't be available to us otherwise. This isn't exactly ground-breaking science – natural chimeras occur all the time, in animals like marmosets for example. In addition to that, scientists have also created chimeric monkeys (with DNA from 6 other monkeys! <http://dgit.in/ChimChimps>) and a mouse with two dads! (<http://dgit.in/2DadMouse>) But human animal chimerism has historically been very hard to accomplish – both due to limitations of research and other external factors such as public backlash and lack of support.

ROADBLOCKS FOR THE VISION

One of the biggest recent examples for the lack of support is the moratorium imposed by the National Institutes of Health (NIH) in USA in September 2015. It cut off all federal funding (not that the taxpayers were bursting to fund it) to research in human animal chimerism while it reviewed the possible ethicality and morality of such research. There was also general public unease over the thought of fusing humans with animals – although people tend to imagine it in a more... dramatic way, no doubt fuelled by science fiction and creepypasta. 'The Island of Dr. Moreau' and 'The Fly' – arguably, one of the best horror films of the last century, both painted vivid pictures of the perils of human animal hybridization. Well, not just fiction is to be blamed either. In the early 1920s, a Russian scientist named Ilya Ivanovich Ivanov tried to create human-ape hybrids. He used female orangutans and chimpanzees and tried to artificially inseminate them with human sperm (his own apparently, but that could just

be a nasty rumour). When that didn't work, he tried it the other way round, trying to inject male chimpanzee sperm into five human female volunteers but was delayed due to the death of his ape donors. By then, ghoulish rumours of his experiments started leaking out (not that it was actually kept secret – he had announced his plans at conventions earlier and actually garnered a little support). The political scene turned against him and he was then exiled to an island where he died shortly after.



Not this kind of Chimera

PIGMAN? HUMIG?

But the winds of change are blowing – this time, on the side of progress. The aforementioned NIH moratorium on funding was lifted on 4th August last year, finally allowing cash strapped labs to start functioning wholly again. Some labs did scrape by with private funding but it wasn't enough to make any major breakthroughs, so the ban lift was much welcomed. But it wasn't all roses – the committee did institute several restrictions. For example, the resulting chimeras NEED to be sterile and the window in which the pluripotent cells can be injected is drastically reduced and monitored. Even so, in the next few months, the scientific community had their first major breakthrough. Salk Institute's Gene Expression Laboratory successfully managed to grow about 186 pig embryos with human cells in it, before terminating them at 4 weeks. Early attempts at injecting human pluripotent cells into the blastocysts didn't work out so well, before they figured out that the human

cells had to mature just a little bit before being injected. After finding the optimum times to make a chimeric embryo involving humans that does survive, scientists are now figuring out how to introduce more human cells into the mix. Right now, they estimate that their most successful attempts have about one human cell in 100,000 pig cells. If these came to term, the resulting pig's organs wouldn't really be usable for a human transplantation because they are still effectively very ...piggy – as in they can't perform their function in a human body. Still, it's a very commendable and significant milestone because now, the question changes from 'if we can grow usable human organs in a pig' to 'when can we grow usable human organs in a pig'.

THE MILLION DOLLAR QUANDARY

There are some darker undertones to the research though. There are plenty of places for exploit and abuse, like perhaps growing a human brain in a pig. Would that result in a sentient pig person? Creating an individual that's its own new species just in the name of science is a terrible thing. Or, would you be okay with eating meat taken from a pig-human hybrid? At what point does the hybrid cease to be a pig and become human? Organisations like PETA are strongly against such research, ostensibly on the grounds of morals and animal abuse. But while perhaps most of us agree that the end doesn't justify the means, medical progress IS needed. The development of medicine isn't always neat and tidy. Most of our anatomical knowledge was first gained from Leonardo Da Vinci's sketches of secretly exhumed corpses. A future where so many people can be saved is worth the discomfort and squeamishness it might evoke in a few people.

Perhaps, that future will be soon here – maybe even in our lifetimes! 

But until that future gets here, people have no choice except to rely on human donors. If you want to help save a life after yours has passed on, you should really consider signing up to be an organ donor. Learn more at <http://dgit.in/OrgnDontrn>